Implementation Of Rail Fence – Row & Column Transformation Technique

AIM:

To write a C program to implement the rail fence transposition technique.

ALGORITHM:

STEP-1: Read the Plain text.

STEP-2: Arrange the plain text in row columnar matrix format.

STEP-3: Now read the keyword depending on the number of columns of the plain text.

STEP-4: Arrange the characters of the keyword in sorted order and the corresponding columns of the plain text.

STEP-5: Read the characters row wise or column wise in the former order to get the cipher text.

PROGRAM:

```
#include <stdio.h>

#include <string.h>


int main()

{

   int i, j, k, l;

   char a[100], c[100], d[100];


   printf("\n\t\tRAIL FENCE CIPHER\n");


   printf("\nEnter the Plain Text: ");     scanf("%s", a);


   l = strlen(a);


   /* Encryption */     j = 0;

   for (i = 0; i < l; i++)
```

```
    {         if (i % 2 == 0)
c[j++] = a[i];
    }

    for (i = 0; i < l; i++)
    {         if (i % 2 != 0)
c[j++] = a[i];
    }      c[j] = '\0';

    printf("\nEncrypted Text : %s\n", c);

    /* Decryption */     if (l %
2 == 0)         k = l / 2;     else
k = (l / 2) + 1;
     j = 0;     for (i = 0; i < k; i++)
    {         d[j] = c[i];         j
+= 2;

    }
     j = 1;     for (i = k; i < l; i++)
    {         d[j] = c[i];         j
+= 2;
    }      d[l] = '\0';

    printf("Decrypted Text : %s\n", d);

    return 0; }
```

OUTPUT:

Output Clear

```
        RAIL FENCE CIPHER

Enter the Plain Text: run

Encrypted Text : rnu
Decrypted Text : run


=== Code Execution Successful ===
```

RESULT:

Thus the rail fence algorithm had been executed successfully